Electromagnetism: Lorentz Force and Maxwell Equations

1. Charged Particles in Combined Electric and Magnetic Fields

A particle with charge q moving with velocity v through an electric field E and magnetic flux density B experiences Lorentz Force:

$$\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$$

2. Differential Formulation of Maxwell's Equations

Gauss's Law for Magnetism: $$\nabla \cdot \mathbf{B} = 0$$
Faraday's Law of Induction: $$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$